

Q3(a)(i)

Maximum EPE stored in the 2 springs $= 2 \times \frac{1}{2} kx^2 = 0.050 \text{ J}$

When this is all transferred to KE,

$$\frac{1}{2}mv_{max}^2 = 0.050$$

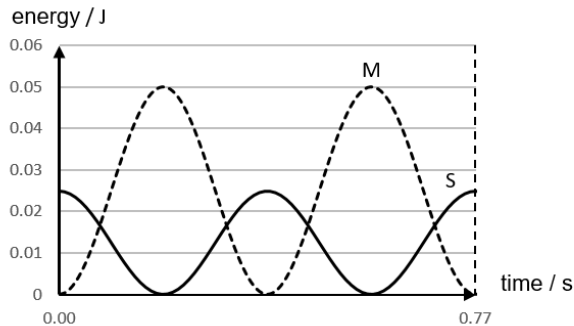
$$v_{max} = \sqrt{\frac{2 \times 0.050}{0.600}} \approx \mathbf{0.41 \text{ m s}^{-1}}$$

Q3(a)(ii)

$$v_{max} = \omega x_0 = (2\pi / T) x_0$$

$$T = \frac{2\pi x_0}{v_{max}} = \frac{2\pi(5 \times 10^{-2})}{0.41} = \mathbf{0.77 \text{ s}}$$

Q3(a)(iii) 1 and 2



Q3(b)

During an earthquake, as the building sways, energy is transferred to the mass-spring system of the damper.

Therefore, amplitude of oscillation of the building decreases, preventing structural damage.

Q4(a)